Review

- Newton's second law $\overrightarrow{F}_{\text{net}} = m_{\text{net}}\vec{a}$ applies to a **single object** or a **collection of objects** ("system")
- All forces (in this class) are pairwise forces. Newton's third law says **pairwise force** are **equal in magnitude and opposite in direction** $F_{A \text{ on } B} = -F_{B \text{ on } A}$
- **Acceleration constraint:** objects connected or that stay in contact move with equal magnitude of acceleration but the direction can be different
- For massless ropes the tension on either end must be equal: $T_{A \text{ on } B} = T_{B \text{ on } A}$, but apply to different objects and may be in different directions
- A massless, frictionless pulley just changes the direction of the force

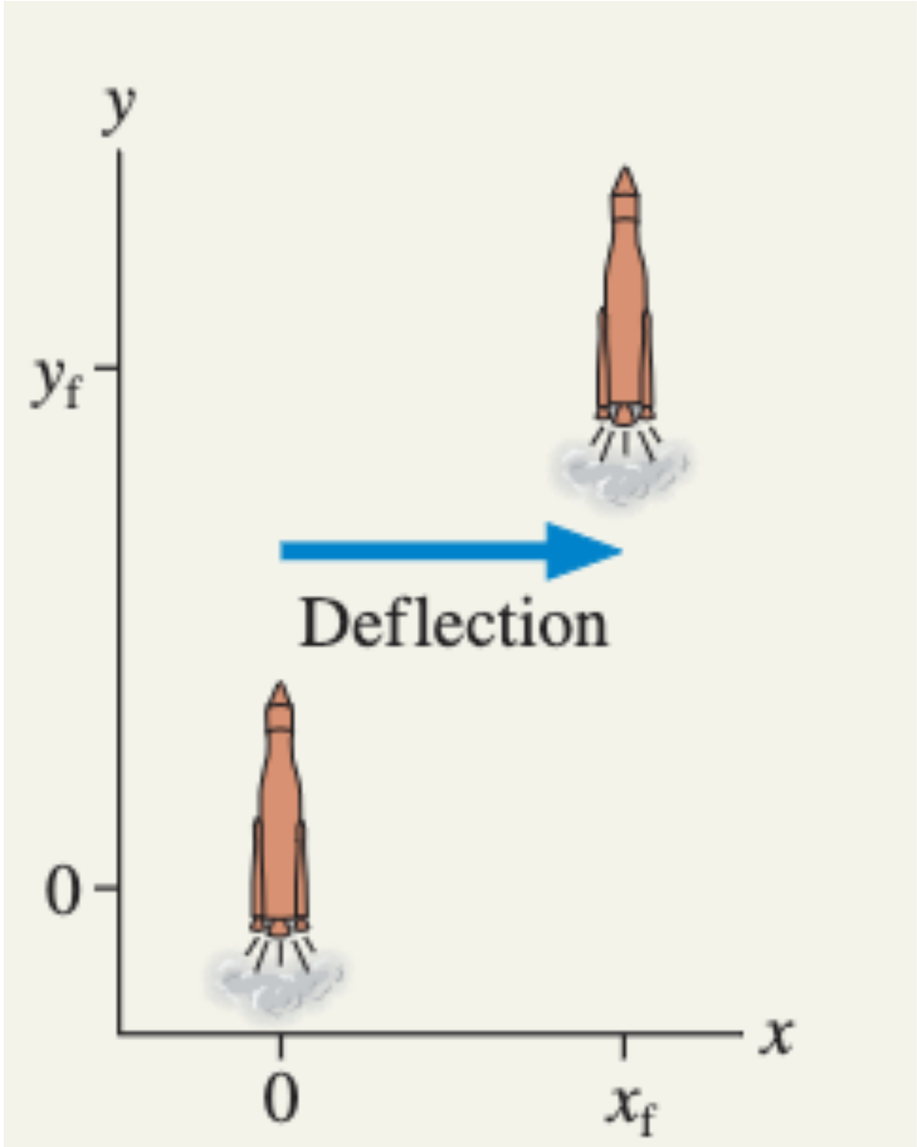
Dynamics and Forces in Two-Dimensions

Dynamics in Two Dimensions

- Newton's second law $\vec{F}_{\text{net}} = m_{\text{net}}\vec{a}$ works the same way in 2D
- Will need to be careful about choice of coordinate system.

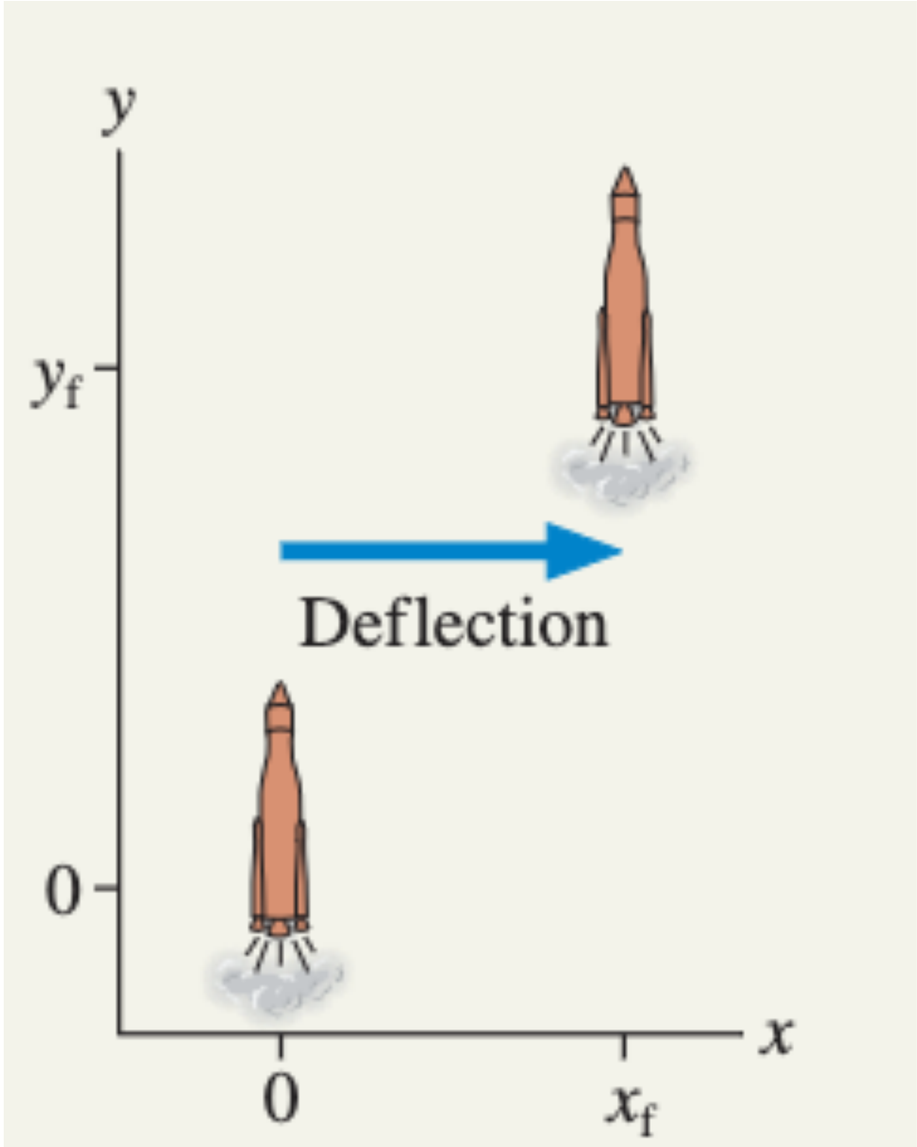
Example

- A small rocket for gathering weather data has a mass of $30\,\mathrm{kg}$ and generates $1500\,\mathrm{N}$ of thrust. On a windy day, the wind exerts a $20\,\mathrm{N}$ horizontal force on the rocket. If the rocket is launched straight up, what is the shape of its trajectory, and by how much has it been deflected sideways when it reaches a height of $1.0\,\mathrm{km}$? Because the rocket goes much higher than this, assume there's no significant mass loss during the first $1.0\,\mathrm{km}$ of flight.



Example

- $y(x) = \dfrac{F_{\text{thrust}} - mg}{F_{\text{wind}}} x$
- $x = \dfrac{F_{\text{wind}}}{F_{\text{thurst}} - mg} y$
- $17\,\text{m}$ deflection at a height of $1000\,\text{m}$

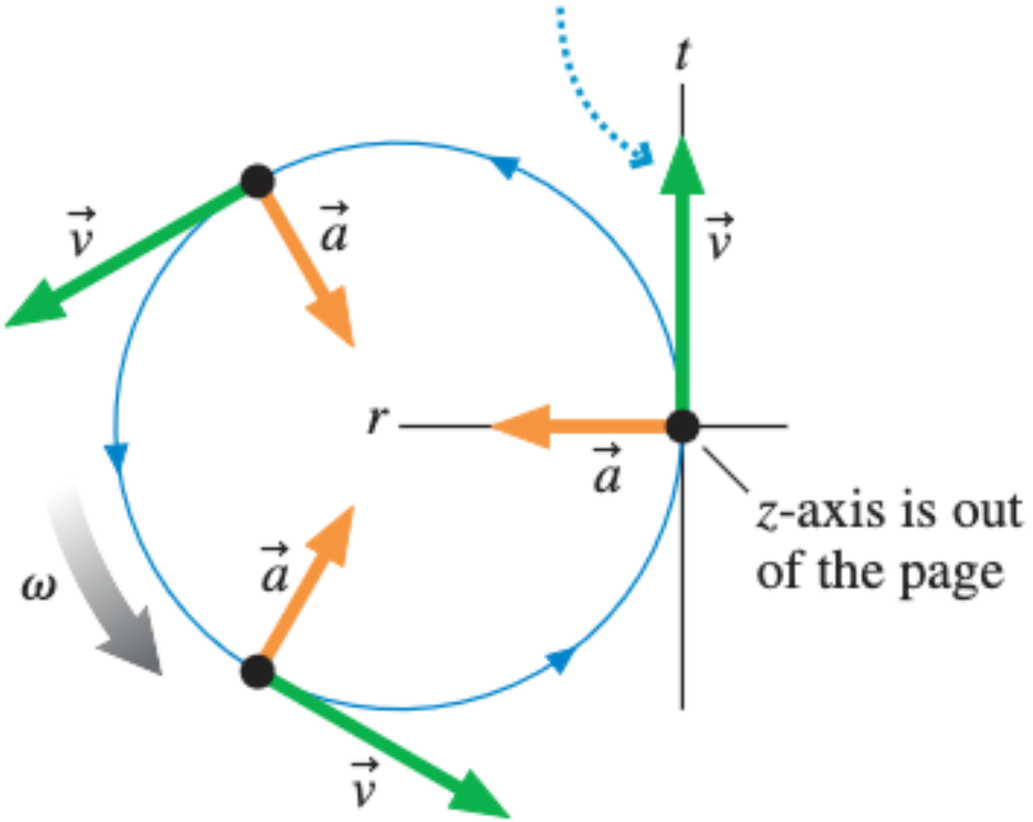


Projectile Motion Revisited

- Consider an object already launched (force that put object in motion no longer in effect)
- Only force acting on the system is gravity: $F_G = w = mg$ (always pointed down)
  - We ignore air resistance and other effects
- From Newton's second law: $\vec{F}_{\text{net}} = m\vec{a} = -mg\hat{j} \rightarrow \vec{a} = -g\hat{j}$ (so $a_x = 0$)

Review: Uniform Circular Motion

- Set up rtz-coordinate system:
  - Origin at particle's position
  - Radial axis: r points from particle towards center of circle (**this is not the radius of the circle**)
  - Tangential axis: t tangent to circle (counter clockwise), (**this is not time**)
  - z-axis is perpendicular to the plan of motion.
- Use angular coordinates
  - θ: angular position
  - ω: angular velocity



Review: Uniform Circular Motion

- Assume the radius of the circular motion does not change (constant)
- ω must be in $\mathbf{rad/s}$
- The (linear) velocity only in the tangent direction
- To keep the object turning acceleration directed towards center (centripetal acceleration)

In the rtz-coordinate system:

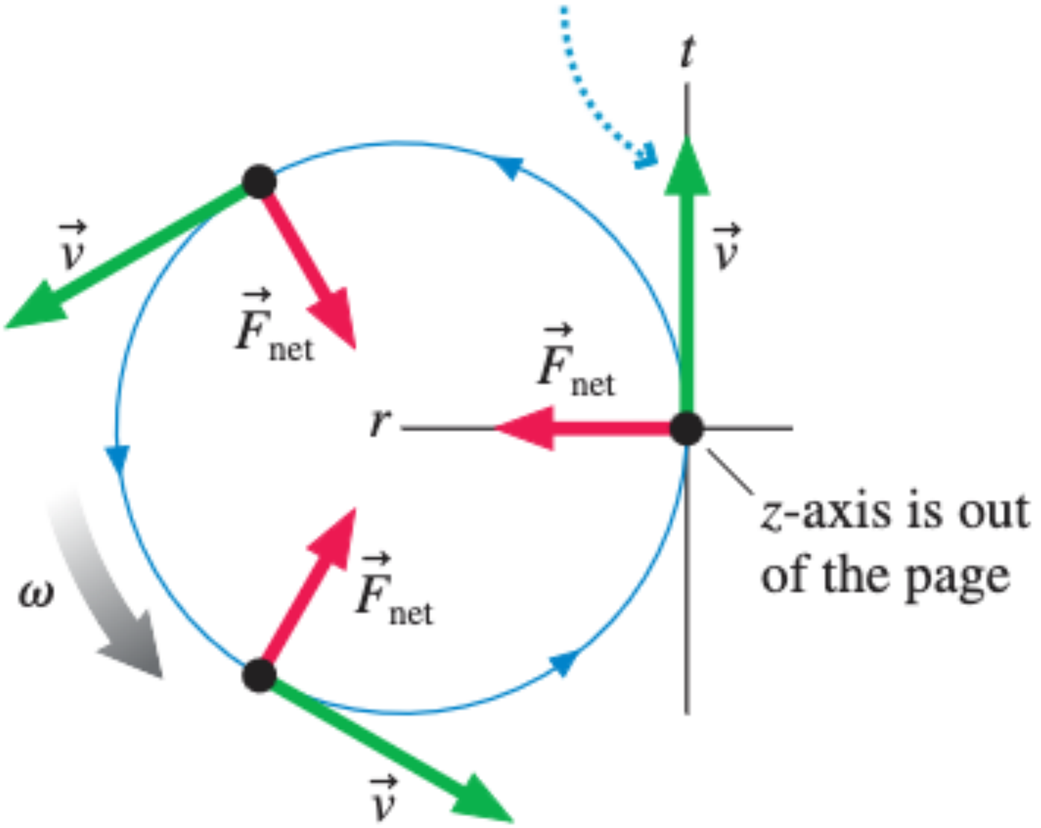
$$v_r = 0 \qquad a_r = \frac{v_t^2}{r} = \omega^2 r$$

$$v_t = \omega r \qquad a_t = 0$$

$$v_z = 0 \qquad a_z = 0$$

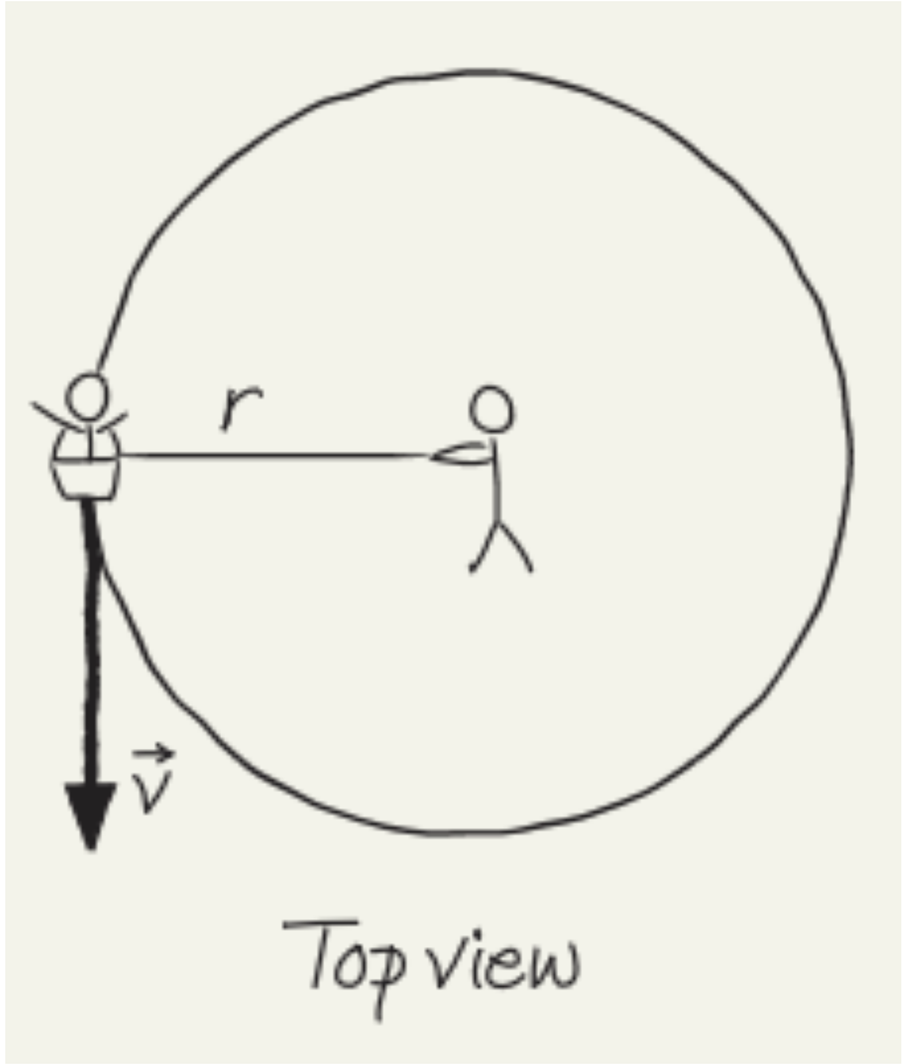
Dynamics of Uniform Circular Motion

- Only acceleration is centripetal acceleration
- By Newton's second law:
  - $F_{\text{net},r} = ma_r = m\dfrac{v_t^2}{r} = m\omega^2 r$
  - $F_{\text{net},t} = ma_t = 0$
  - $F_{\text{net},z} = ma_z = 0$



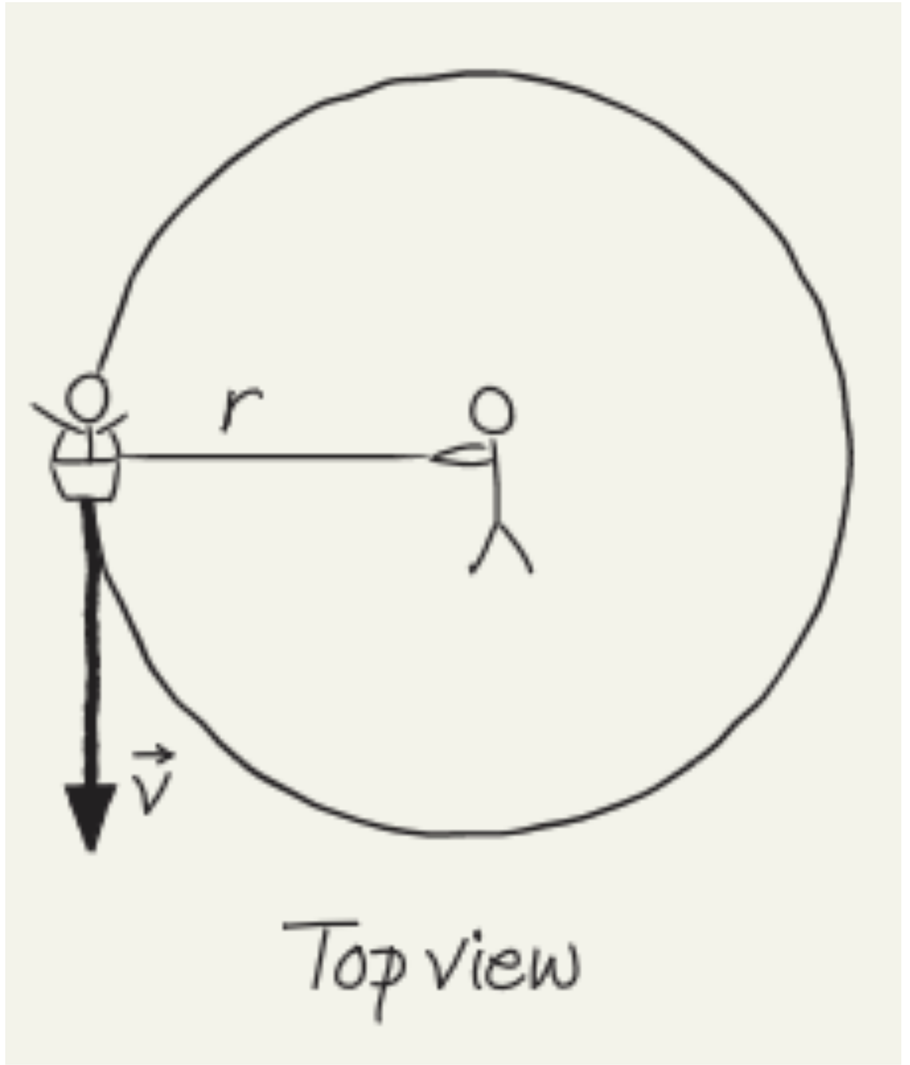
Example

- An energetic father places his $20\,\mathrm{kg}$ child on a $5\,\mathrm{kg}$ cart to which a $2.0\,\mathrm{m}$ long rope is attached. He then holds the end of the rope and spins the cart and child around a circle, keeping the rope parallel to the ground. If the tension in the rope is $100\,\mathrm{N}$, how many revolutions p[er min (rpm) does the cart make? Rolling friction between the cart's wheel and the ground is negligible.



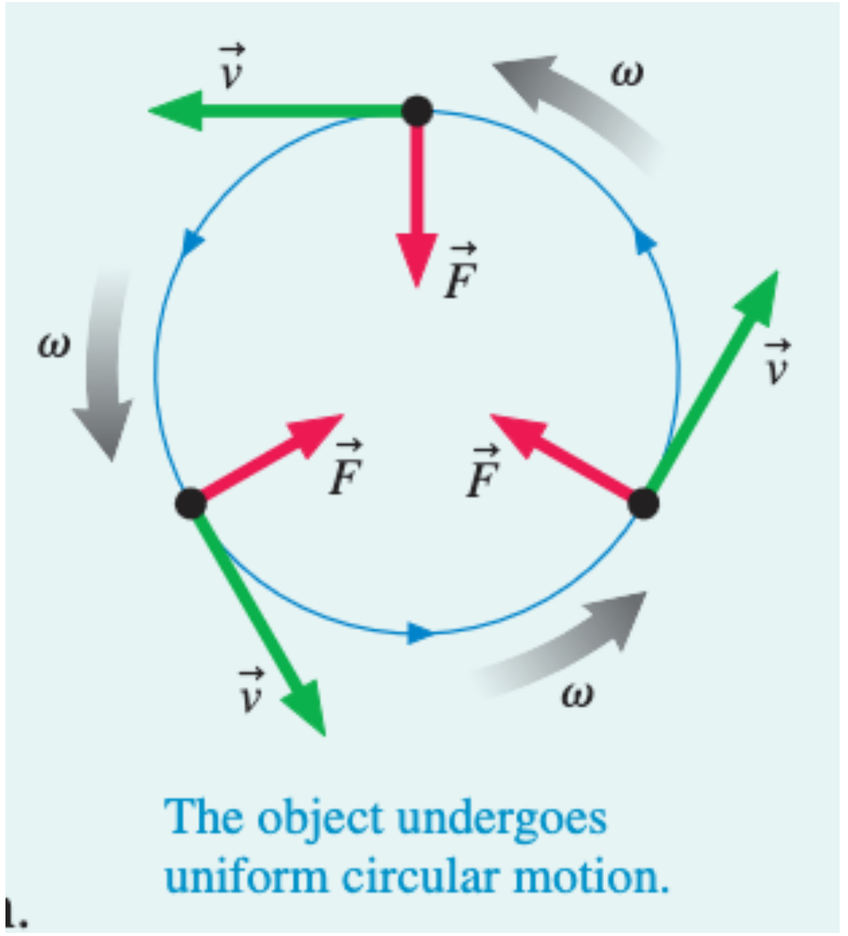
Example

- $F_{\text{net},r} = ma_r = m\omega^2 r = T \rightarrow \omega = \sqrt{\frac{T}{m\,r}}$
- $\sim 14\,\text{N}$



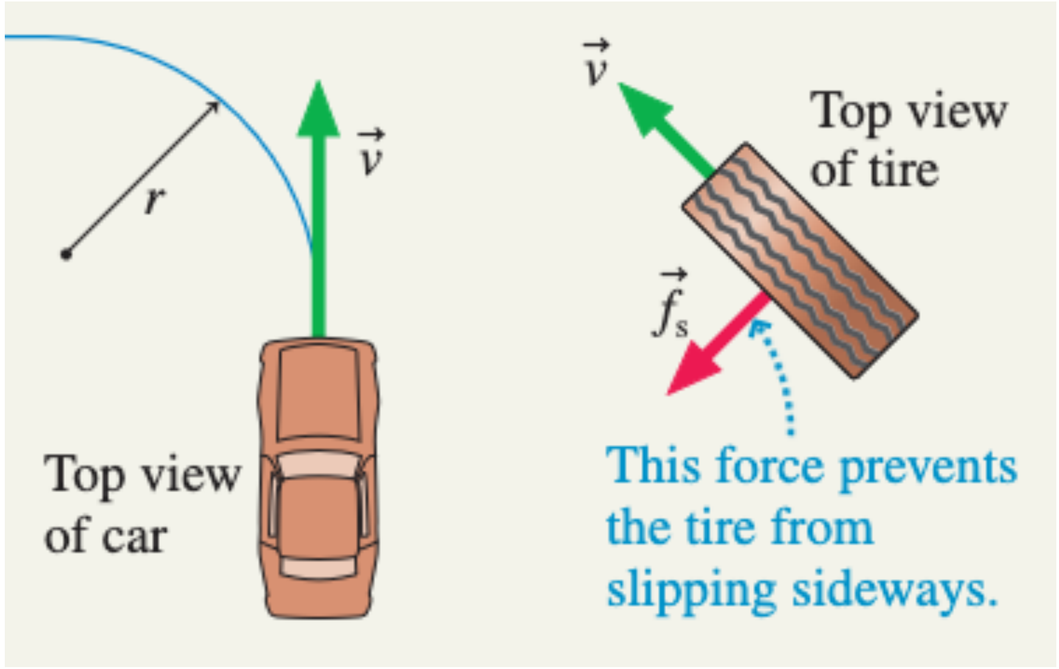
Central Force with Constant r Model

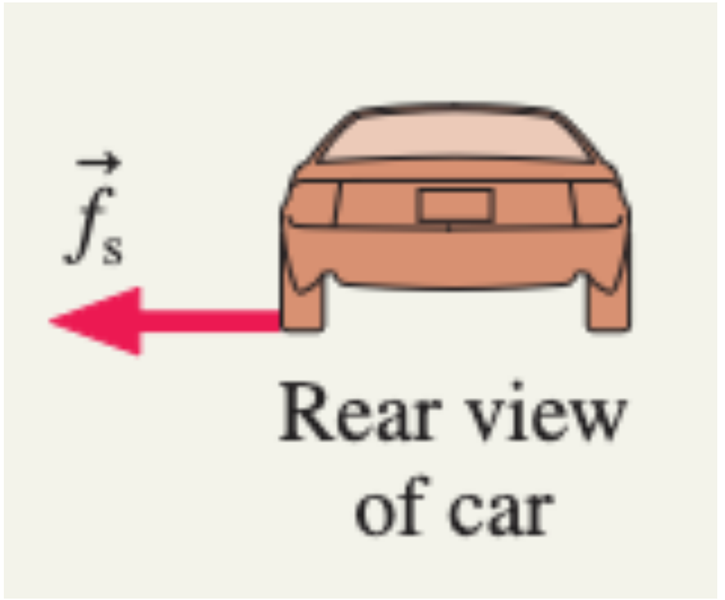
- Object for which the net force is constant in magnitude but points towards the center
- The force causes centripetal acceleration and uniform circular motion
- $\overrightarrow{F}_{\text{net}} = m\frac{v_t^2}{r}\hat{r} = m\omega^2 r\hat{r}$
- Use kinematics of uniform circular motion



Example

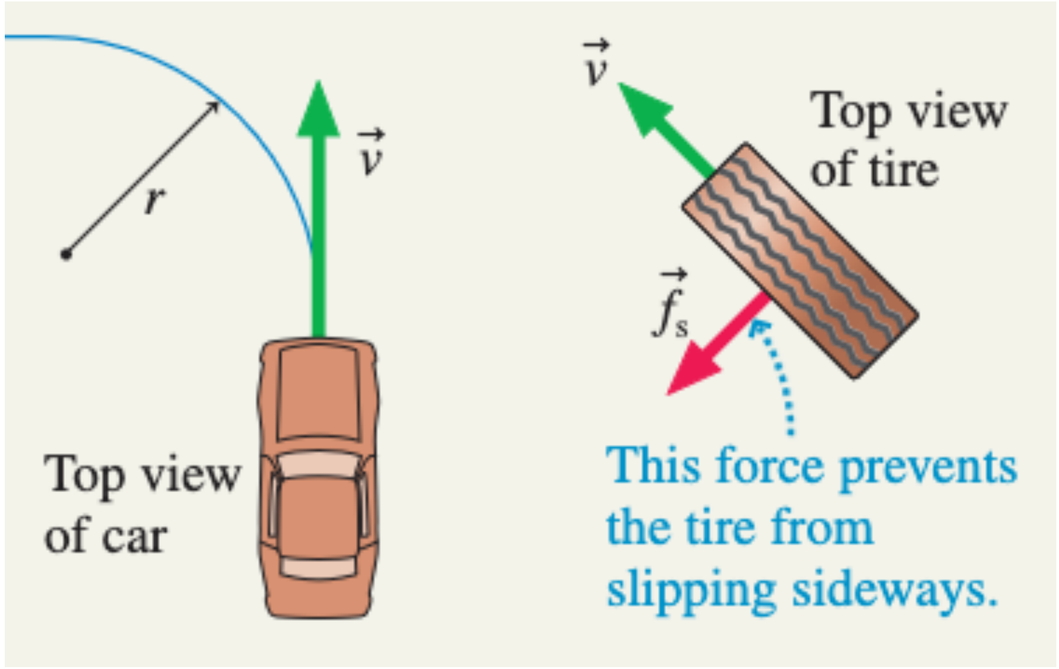
- What is the maximum speed with which a $1500\,\mathrm{kg}$ car can make a left turn around a curve of radius $50\,\mathrm{m}$ on a level (unbanked) road without sliding?

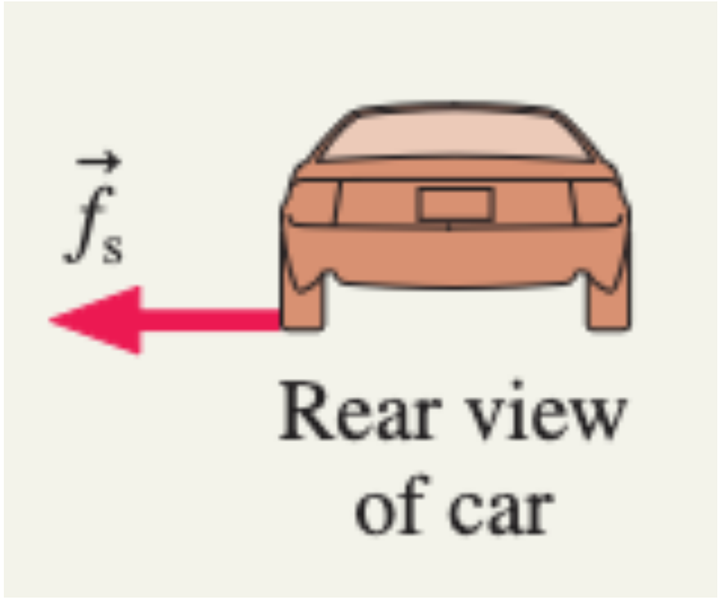




Example

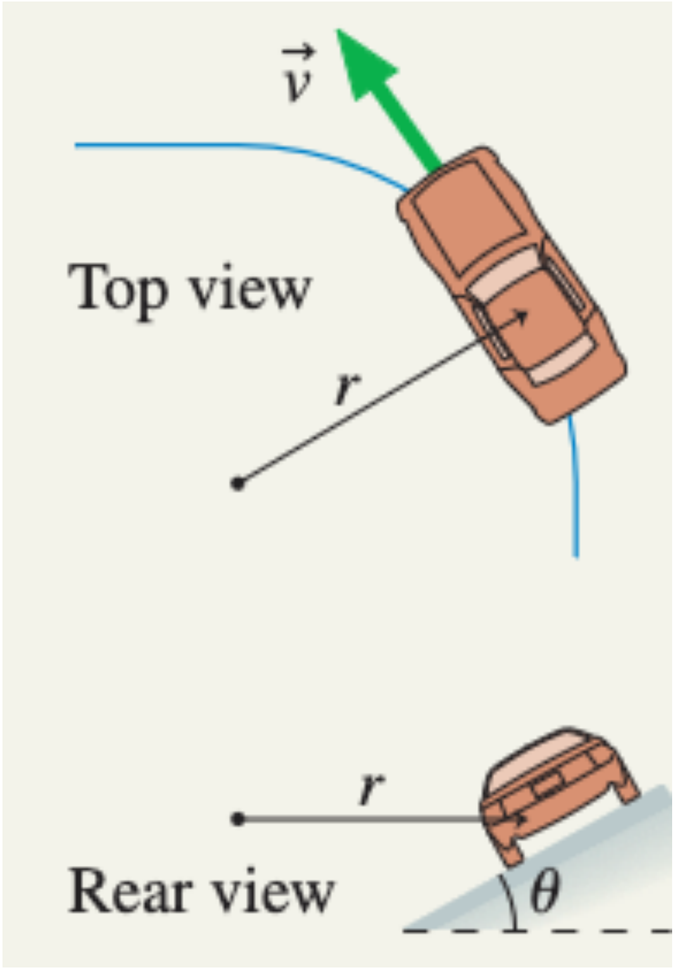
- $v_{\max} = \sqrt{\dfrac{rf_{s,\max}}{m}} = \sqrt{\mu_s rg} = 22\ \text{m/s}$





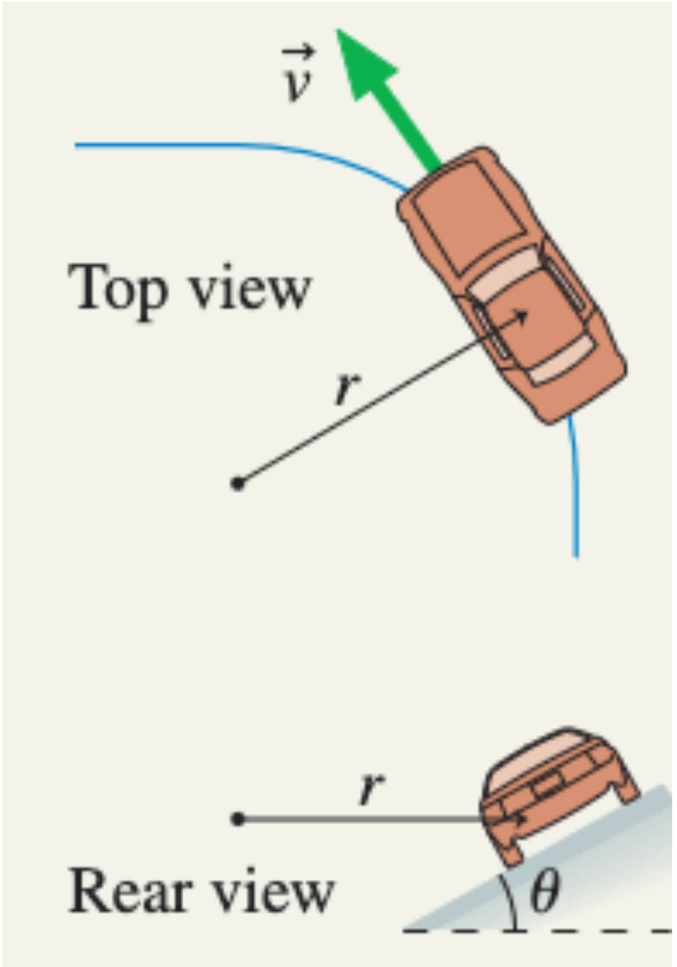
Example

- A highway curve of radius $70\,\mathrm{m}$ is banked at $15°$ angle. At what speed v_0 can a car take this curve without assistance from friction?



Example

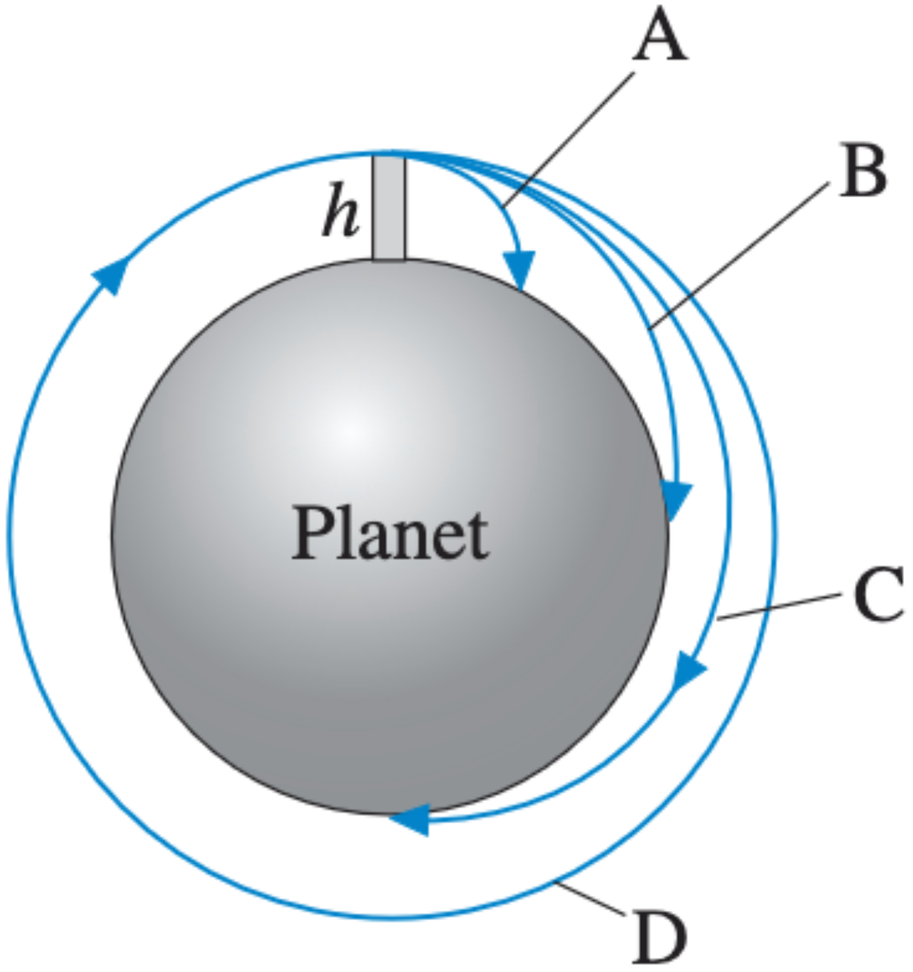
- Newton's second law
  - $F_{\text{net},r} = n \sin\theta = m\dfrac{v_0^2}{r}$
  - $F_{\text{net},z} = n\cos\theta - mg = 0$
- $n = \dfrac{mg}{\cos\theta}$
- $v_0 = \sqrt{rg\tan\theta} = 14\ \text{m/s}$



Circular Orbits

- Start with projectile motion
- At some height h, launch object with velocity v_0 horizontally
- Newton's second law:

$$F_{\text{net},y} = ma_y = F_G = -mg$$

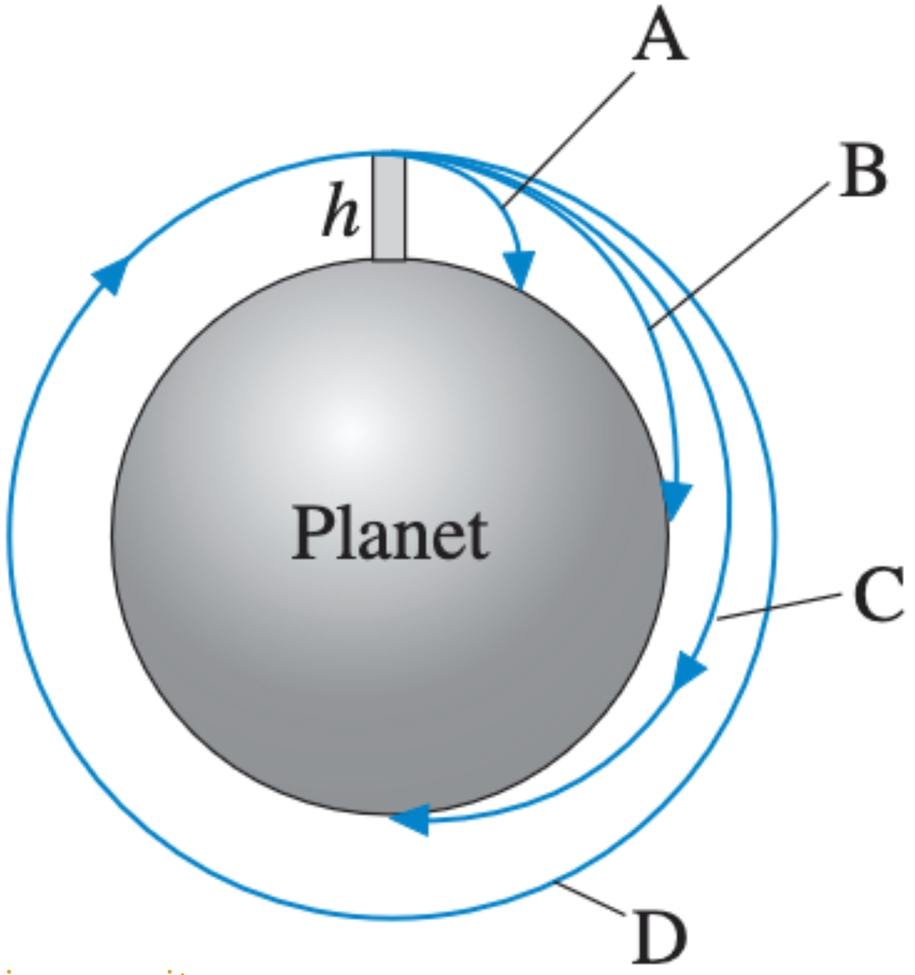


Circular Orbits

- Increase v_0 until sufficient large that object's trajectory and curve of planet are parallel (**orbit**)
- Use rtz-coordinates
- Use Newton's second law for uniform circular motion
- Newton's second law:

$$F_{\text{net},r} = m\frac{v_c^2}{R} = F_G = mg$$

- $v_c = \sqrt{Rg}$



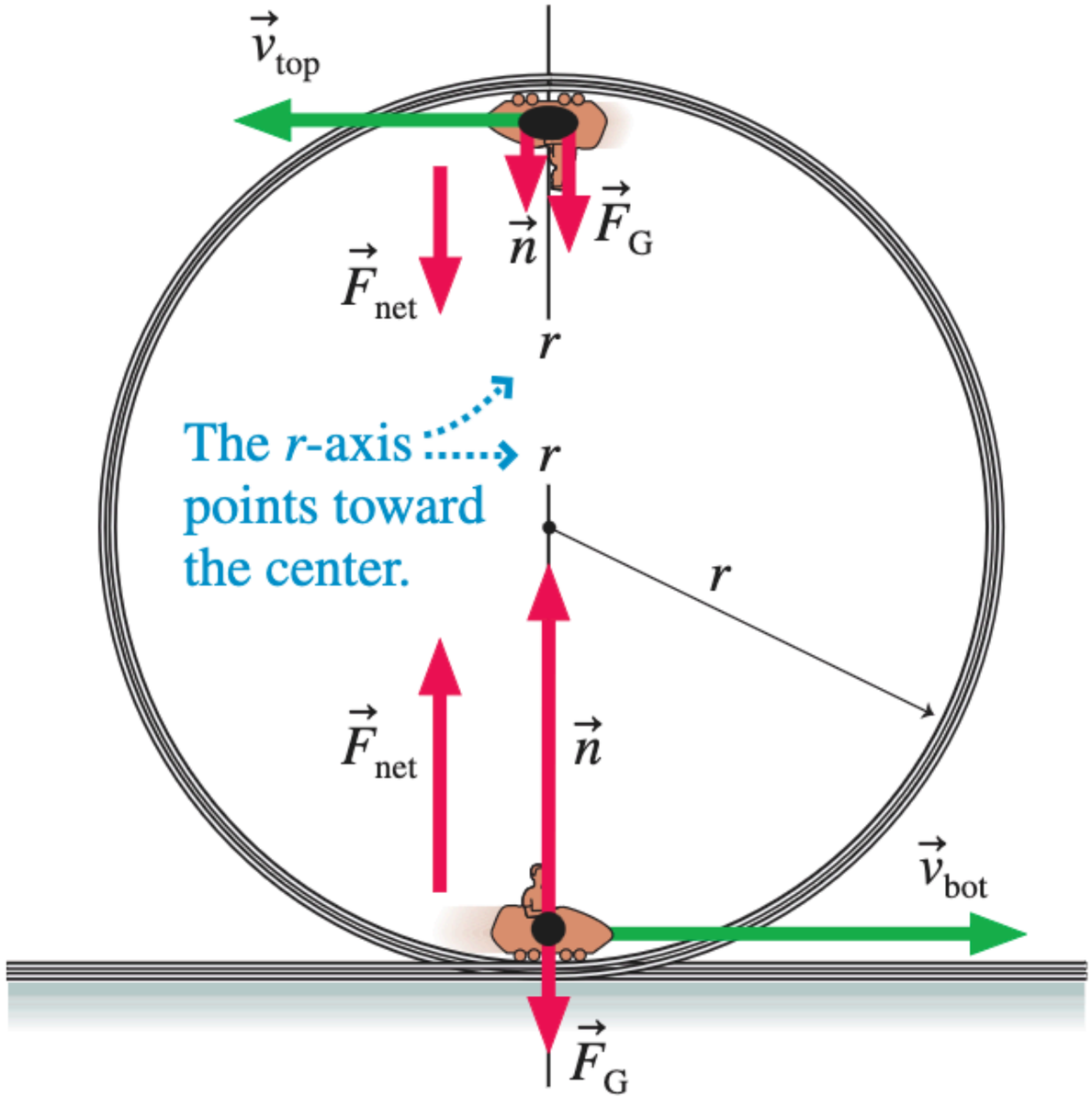
Only force acting on object is gravity, therefore orbiting is object constantly falling under influence of gravity

- Consider swinging a bucket of water or roller coaster doing a loop-the-loop
- Motion not uniform circular motion (car slows down as it goes up one side and speeds up as it comes down the other)
- At the very top and bottom points, only direction changes, not speed, so acceleration purely centripetal
- Newton's second law:

$$F_{\text{net},r} = m\frac{v_t^2}{R} = n_r \pm mg$$

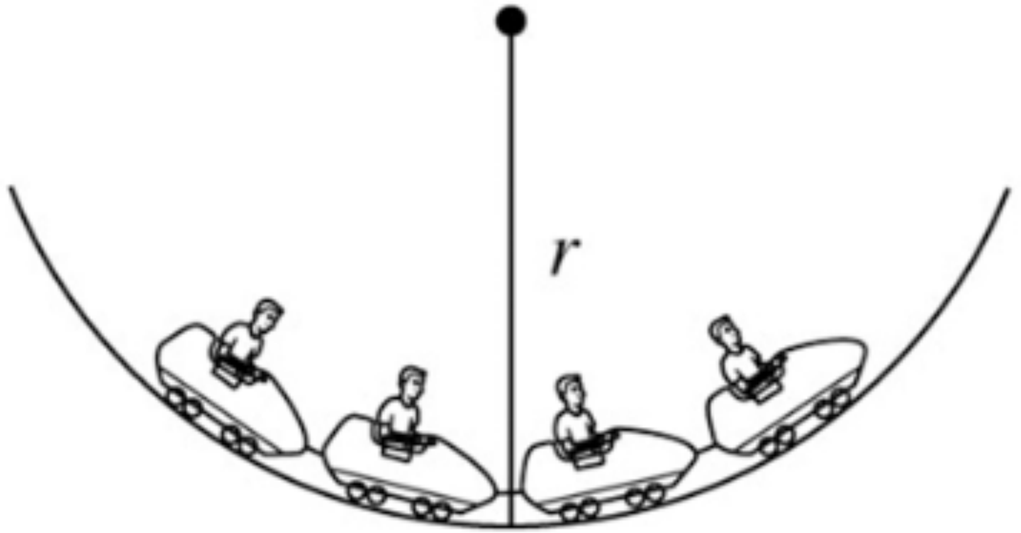
$$n = m\frac{v_{\text{top}}^2}{R} - mg$$

$$v_c = \sqrt{Rg}$$



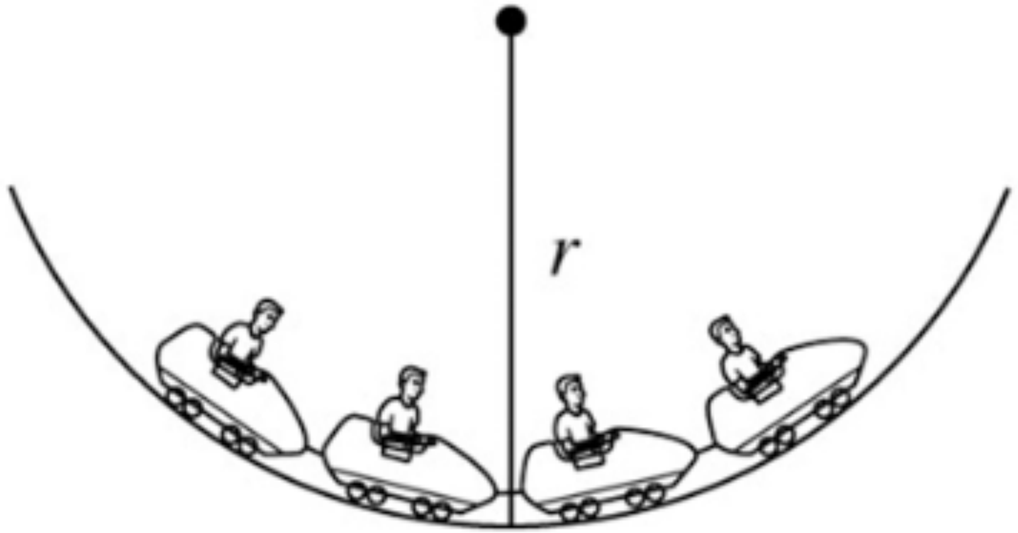
Example

- The weight of passengers on a roller coaster increase by $50\,\%$ as the car goes through a dip with a $30\,\mathrm{m}$ radius of curvature. What is the car's speed at the bottom of the dip?



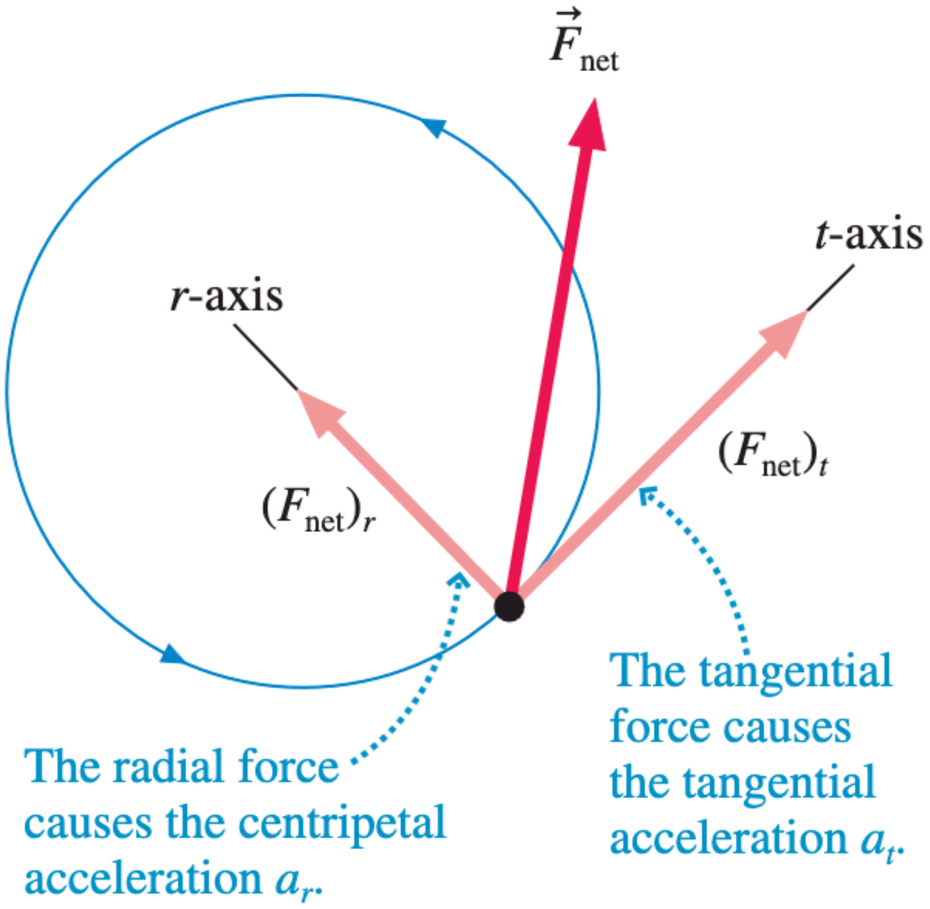
Example

- The weight of passengers on a roller coaster increase by $50\,\%$ as the car goes through a dip with a $30\,\mathrm{m}$ radius of curvature. What is the car's speed at the bottom of the dip?
- $12\,\mathrm{m/s}$



Review: Nonuniform Circular Motion

- Using radial-tangent-z (rtz) coordinate system
- Coordinates:
  - Angular position: θ
  - Angular velocity: ω (in radians)
  - Angular acceleration: α



Review: Nonuniform Circular Motion

- Assume radius of circular motion does not change (constant)
- Object's tangential velocity changing (equivalently, its angular velocity is changing)
- Velocity in the tangent direction
- Motion affected by centripetal and tangental acceleration

In the *rtz*-coordinate system:

$$v_r = 0 \qquad a_r = \frac{v_t^2}{r} = \omega^2 r$$

$$v_t = \omega r \qquad a_t = \frac{dv_t}{dt} = \alpha r$$

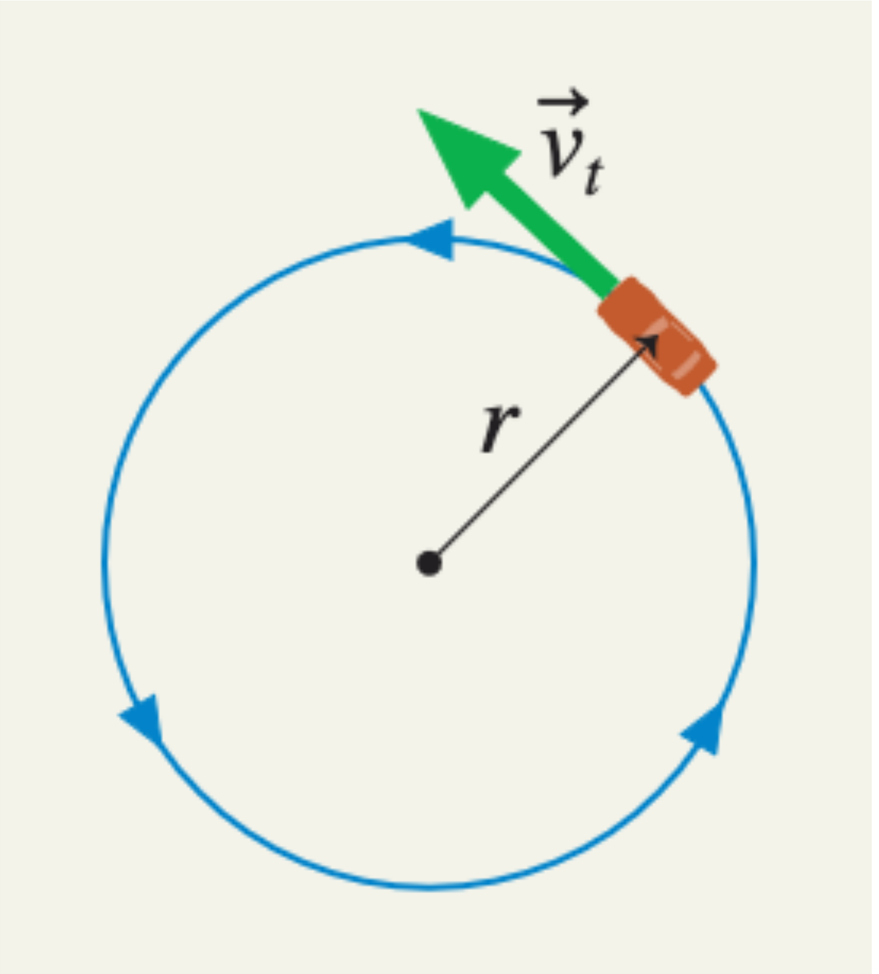
$$v_z = 0 \qquad a_z = 0$$

Dynamics of Nonuniform Circular Motion

- Newton's second law
  - $F_{\text{net},r} = ma_r = m\dfrac{v_t^2}{r} = m\omega^2 r$
  - $F_{\text{net},t} = ma_t$
  - $F_{\text{net},z} = 0$

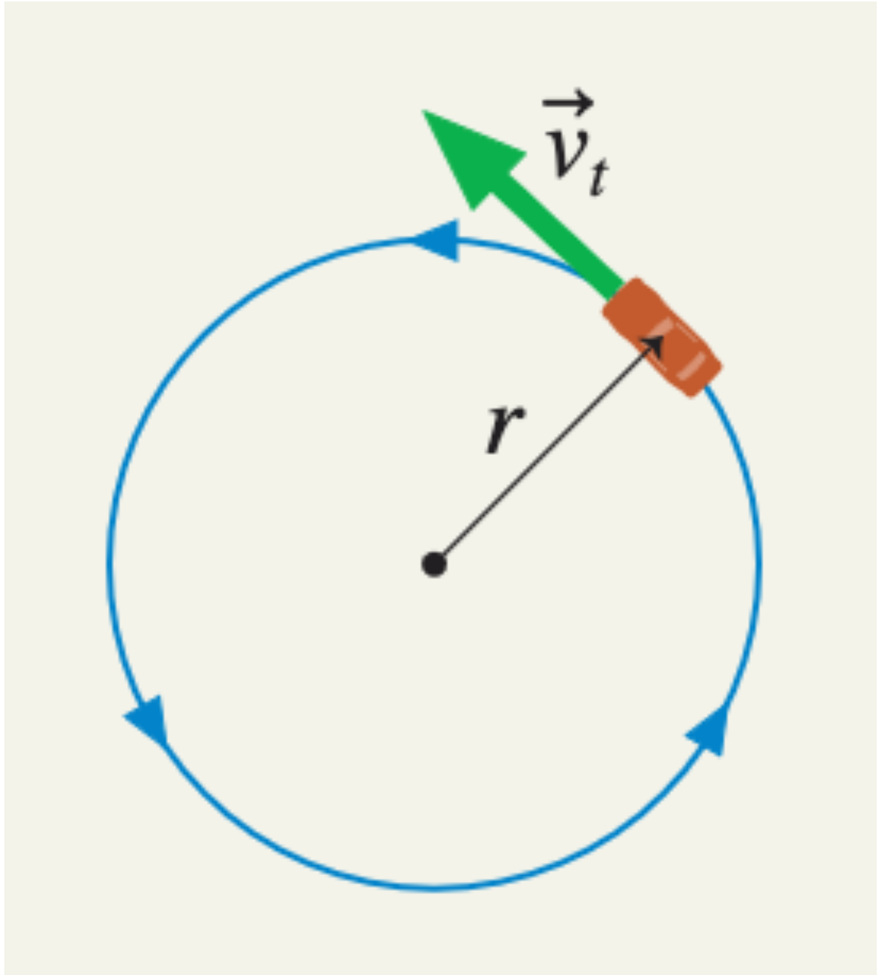
Example

- A $1500\,\mathrm{kg}$ car drives around a flat, $50\,\mathrm{m}$ diameter track, starting from rest. The drive wheels supply a small but steady $525\,\mathrm{N}$ force in the forward direction. The coefficient of static friction between the car tires and the road is 0.90. How many revolutions of the track have been made when the car slides out of the curve?



Example

- $v_{t,\max}^2 = \mu_s R g$
- $\Delta\theta_{\max} = \dfrac{\mu m g}{2F_{\text{wheel}}}$
- 2.0 rev



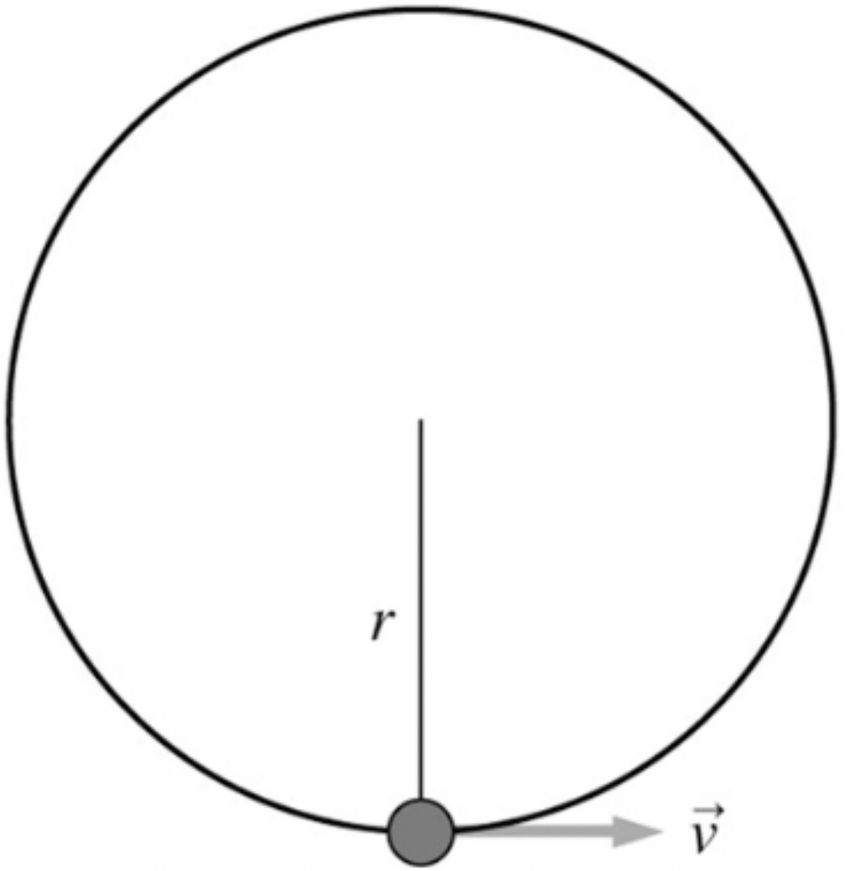
Problem Solving Strategy

Circular- motion problems

- Model the object as a particle and make other simplifying assumptions.
- Draw a picture (use rtz-coordinates with the r-axis pointing towards the center of the circle)
- Define symbols and identify what the problem is trying to find.
- Identify the forces and show them on a free-body diagram
- Solve: Newton's second law is
  - $F_{\text{net},r} = ma_r = m\dfrac{v_t^2}{r} = m\omega^2 r$
  - $F_{\text{net},t} = ma_t$ (for uniform circular motion $a_t = 0$)
  - $F_{\text{net},z} = 0$

Example

- A $600\ \mathrm{g}$ ball swings in a vertical circle at the end of a $1.1\ \mathrm{m}$ long string. When the ball is at the bottom of the circle, the tension in the string is $18\ \mathrm{N}$. What is the speed of the ball at that point?



Example

- A $600\ \mathrm{g}$ ball swings in a vertical circle at the end of a $1.1\ \mathrm{m}$ long string. When the ball is at the bottom of the circle, the tension in the string is $18\ \mathrm{N}$. What is the speed of the ball at that point?

- $5.5\ \mathrm{m/s}$

